

1. Module-Level Design

1.1 Retrieval Module

- Fetches relevant information from knowledge base
- Input: user query
- Output: matched context

1.2 Small Talk Module

- Handles greetings and casual queries
- Examples: hello, hi, how are you

1.3 Intent Detection Module

- Classifies queries into:
- FAQ
- Escalation
- Unknown

1.4 Confidence Module

- Assigns confidence score
- Context found → 0.9
- No context → 0.2

1.5 Processing Module

- Combines query, context, intent, confidence
- Decides next step

1.6 Routing Module

- Routes query to:
- Response generation
- Escalation

1.7 HITL Module

- Handles escalation
- Triggered when:
- No data
- Low confidence
- User requests human

1.8 Output Module

- Returns final answer
- Includes status

2. Data Structures

2.1 Graph State

```
{  
  "query": "string",  
  "context": "string",  
  "answer": "string",  
  "escalate": "boolean"  
}
```

2.2 Knowledge Base

```
{  
  "refund": "...",  
  "warranty": "...",  
  "shipping": "..."  
}
```

2.3 Query-Response Schema

```
{  
  "query": "string",  
  "answer": "string",  
  "status": "resolved/escalated"  
}
```

3. Workflow Design (LangGraph)

Nodes:

- retrieve_context
- process_query
- generate_output

Flow:

retrieve → process → output

4. Conditional Routing Logic

Escalation triggered when:

- User asks for human
- Confidence < 0.5
- No context found

5. HITL Design

Flow:

1. Detect issue
2. Mark escalation
3. Return:
 “Escalated to human agent”

6. API / Interface Design

Input:

```
{  
  "query": "What is refund policy?"  
}
```

Output:

```
{  
  "answer": "Refunds are allowed within 7 days",  
  "status": "resolved"  
}
```

7. Error Handling

- Empty query → prompt user
- No context → escalate
- System failure → fallback

8. Execution Flow

1. User enters query
2. Retrieval module finds context
3. Processing module evaluates
4. Routing decision made
5. Output returned

9. Design Benefits

- Modular design
- Easy to extend
- Scalable system
- Clear workflow
- Intelligent decision-making